Customer Guide: Install NGINX Plus Ingress for Kubernetes DNS

This guide installs only the F5 NGINX Ingress Controller with NGINX Plus. It assumes Kubernetes is already running and the DNS service is already deployed.

The goal is to expose an existing Kubernetes DNS service on TCP and UDP port 53 using F5 NGINX ‘GlobalConfiguration‘ and ‘TransportServer‘ resources.

Assumptions

- Kubernetes cluster is ready.
- ‘kubectl‘ is configured for the target cluster.
- Helm is installed.
- The DNS backend service already exists.
- The customer has a valid NGINX Plus JWT license token.
- The cluster nodes can pull from ‘private-registry.nginx.com‘.

Example backend used in this guide:

```
DNS namespace: dns-lab
DNS service:   coredns
DNS port:      53
```

If the customer DNS service has different names, replace:

```
dns-lab/coredns:53
```

with:

```
<dns-namespace>/<dns-service>:<dns-port>
```

1. Verify the Existing DNS Service

```
kubectl -n dns-lab get svc coredns
kubectl -n dns-lab get endpoints coredns
```

The service must expose both protocols:

```
53/TCP
53/UDP
```

2. Create NGINX Plus Secrets

Save the NGINX Plus JWT locally as:

```
license.jwt
```

Create the NGINX namespace:

```
kubectl create namespace nginx-ingress --dry-run=client -o yaml | kubectl apply -f -
```

Create the NGINX Plus license secret:

```
kubectl -n nginx-ingress create secret generic nplus-license \
  --from-file=license.jwt=./license.jwt \
  --type=nginx.com/license
```

Create the private registry pull secret:

```
kubectl -n nginx-ingress create secret docker-registry regcred \
  --docker-server=private-registry.nginx.com \
  --docker-username="$(cat ./license.jwt)" \
  --docker-password=none
```

Do not commit or share ‘license.jwt‘.

3. Create Helm Values

Create this file:

```
nginx-plus-dns-values.yaml
```

Use this content:

```
controller:
  kind: daemonset
  nginxplus: true
  hostNetwork: true
  dnsPolicy: ClusterFirstWithHostNet

  image:
    repository: private-registry.nginx.com/nginx-ic/nginx-plus-ingress
    tag: 5.4.1

  serviceAccount:
    imagePullSecretName: regcred

  mgmt:
    licenseTokenSecretName: nplus-license

  enableCustomResources: true

  globalConfiguration:
    create: true
    spec:
      listeners:
      - name: dns-tcp
        port: 53
        protocol: TCP
      - name: dns-udp
        port: 53
        protocol: UDP

  customPorts:
  - name: dns-tcp
    containerPort: 53
    protocol: TCP
  - name: dns-udp
    containerPort: 53
    protocol: UDP

  service:
    create: false
```

Notes:

- ‘hostNetwork: true‘ binds NGINX directly on the Kubernetes node network.
- Port ‘53‘ must be free on the nodes where this DaemonSet runs.
- If another DNS service is already listening on node port ‘53‘, this deployment will fail.

4. Install NGINX Plus Ingress Controller

```
helm upgrade --install nginx-ingress oci://ghcr.io/nginx/charts/nginx-ingress \
  --version 2.5.1 \
  --namespace nginx-ingress \
  -f nginx-plus-dns-values.yaml
```

Wait for the controller:

```
kubectl -n nginx-ingress wait \
  --for=condition=Ready pod \
  -l app.kubernetes.io/name=nginx-ingress \
  --timeout=600s
```

Verify:

```
kubectl -n nginx-ingress get pods
kubectl -n nginx-ingress get globalconfiguration
helm -n nginx-ingress list
```

5. Create TransportServer Resources

Create this file:

```
dns-transportservers.yaml
```

Use this content:

```
apiVersion: k8s.nginx.org/v1
kind: TransportServer
metadata:
  name: dns-tcp
  namespace: dns-lab
spec:
  listener:
    name: dns-tcp
    protocol: TCP
  upstreams:
  - name: dns
    service: coredns
    port: 53
  action:
    pass: dns
---
apiVersion: k8s.nginx.org/v1
kind: TransportServer
metadata:
  name: dns-udp
  namespace: dns-lab
spec:
  listener:
    name: dns-udp
    protocol: UDP
  upstreams:
```

```
- name: dns
  service: coredns
  port: 53
upstreamParameters:
  udpRequests: 1
  udpResponses: 1
action:
  pass: dns
```

Apply it:

```
kubectl apply -f dns-transportservers.yaml
```

6. Verify TransportServer Status

```
kubectl -n dns-lab get transportserver
kubectl -n dns-lab describe transportserver dns-tcp
kubectl -n dns-lab describe transportserver dns-udp
```

Expected status:

```
NAME      STATE   REASON
dns-tcp   Valid   AddedOrUpdated
dns-udp   Valid   AddedOrUpdated
```

7. Test DNS

Set the node IP where NGINX Plus is running:

```
NODE_IP=<nginx-ingress-node-ip>
```

Test UDP:

```
dig @"$NODE_IP" -p 53 app.example.local +short
dig @"$NODE_IP" -p 53 api.example.local +short
```

Test TCP:

```
dig @"$NODE_IP" -p 53 app.example.local +tcp +short
dig @"$NODE_IP" -p 53 api.example.local +tcp +short
```

Expected demo answers:

```
app.example.local -> 10.10.10.10
api.example.local -> 10.10.10.20
```

8. Troubleshooting

Check image pull and license issues:

```
kubectl -n nginx-ingress get events --sort-by=.lastTimestamp
kubectl -n nginx-ingress describe pod -l app.kubernetes.io/name=nginx-ingress
kubectl -n nginx-ingress get secret nplus-license regcred
```

Check NGINX Plus logs:

```
kubectl -n nginx-ingress logs -l app.kubernetes.io/name=nginx-ingress --tail=100
```

Check listener configuration:

```
kubectl -n nginx-ingress get globalconfiguration nginx-ingress-controller -o yaml
```

Check TransportServer configuration:

```
kubectl -n dns-lab get transportserver -o yaml
```

Check the DNS backend:

```
kubectl -n dns-lab get pods
kubectl -n dns-lab get svc coredns
kubectl -n dns-lab get endpoints coredns
```

Common issues:

- Port '53' is already used on the node.
- The NGINX Plus JWT is expired or invalid.
- The cluster cannot reach 'private-registry.nginx.com'.
- The DNS service name or namespace in the TransportServer is wrong.
- The DNS service does not expose both TCP and UDP port '53'.

References

- F5 NGINX Ingress Controller Helm install:
https://docs.nginx.com/nginx-ingress-controller/install/helm/

- F5 NGINX TransportServer resources:
https://docs.nginx.com/nginx-ingress-controller/configuration/transportserver-resource/

- F5 NGINX GlobalConfiguration resource:
https://docs.nginx.com/nginx-ingress-controller/configuration/global-configuration/globalconfiguration-resource/